Results outline

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Research questions

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
4. How did growth change across years, and how did it relate to climate?

Results

1. Summary basic information
   (a) Wildchrokie includes 81 individuals of four deciduous tree species from four sites with leaf phenology data spaning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS).
   (b) GSL ranged from 90 days (*B. alleghaniensis* in 2018) to 173 days (*A. incana* in 2018)
   (c) GS thermal sum ranged from 1465.1 GDD (*B. alleghaniensis* in 2018) to 2513.6 GDD, (*A. incana* in 2018)
   (d) The earliest leafout day of year was 112 (2019) and the latest, 157 (2019). The earliest budset day of year was 225 (2018) and the latest, 297 (2018)
2. Does tree growth increase with longer growing seasons?
   (a) Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it.
   (b) Longer thermal season increased growth for *B. alleghaniensis* ($\beta = 0.13$, UI_{90} [0.01, 0.25]), *B. papyrifera* ($\beta = 0.44$, UI_{90} [0.17, 0.71]), and *B. populifolia* ($\beta = 0.21$, UI_{90} [0.07, 0.35]), but *A. incana* trended in the same direction ($\beta = 0.07$, UI_{90} [-0.03, 0.17]) (Figure 1a and e) (TableS1).
   (c) Only one out of those three species also grew more with longer calendar season (*B. papyrifera*: $\beta = 0.18$, UI_{90} [0.06, 0.32]), but an additional species that didn't grow more with warmer thermal season, did with longer calendar season (*A. incana*: $\beta = 0.05$, UI_{90} [0, 0.1]).
   (d) *B. alleghaniensis* ($\beta = 0.03$, UI_{90} [-0.03, 0.09]), and *B. populifolia* ($\beta = 0.06$, UI_{90} [-0.02, 0.13]) trended in the same direction (Figure 1b and f) (Table S1).
3. Did an later start and end of season increase growth?
   (a) A longer calendar season increased growth for two species, but neither grew more with a later start of season.
   (b) *A. incana* grew more with longer calendar season, but grew less with a later start of season (β = -0.07, UI_{90} [-0.14, -0.01]) (Figure 1c and g) (Table S1).
   (c) *B. alleghaniensis* which did not grow more with longer calendar season, did increase growth with a later start of season ($\beta = 0.1$, UI_{90} [0, 0.2]) (Figure 1c and g) (Table S1).
   (d) Three species grew more with a later end of season (*B. alleghaniensis*: $\beta = 0.12$, UI_{90} [0.05, 0.18], *B. populifolia*: $\beta = 0.1$, UI_{90} [0.03, 0.16]), with *B. papyrifera* ($\beta = 0.21$, UI_{90} [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 1d and h) (Table S1).

4. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
    (a) *B. papyrifera* grew the least (α = -3.76, UI_{90} [-8.62, 1.12]) (Figure S3), but was the most responsive to longer growing seasons (Figure 1a and b).
    (b) *A. incana* grew the most (α = 1.02, UI_{90} [-3.07, 5.2]) (Figure S3), but had contrasting responses to longer growing seasons (Figure 1a and b).
    (c) Growth did not change across any provenances (Figure 2) (Table S2), but there was slightly more variation across sites ($\sigma = 0.18$, UI_{90} [0.02, 0.5) than across treeids ($\sigma = 0.17$, UI_{90} [0.05, 0.27]
5. How did growth change across years, and how did it relate to climate?
    (a) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018) (Figure S5).
    (b) Growth was highest in 2018 (0.18, UI_{90} [-0.74, 1.12]) (Figure 3), but it was neither the wettest, the driest, the shortest, the latest SOS, the earliest EOS nor the highest thermal sum, unlike our finding that warmer and longer seasons increase growth for most of our species (Figure S1 and Figure S2).
    (c) Growth was lowest in 2020 (-0.44, UI_{90} [-1.36, 0.5]) (Figure 3) and that year was the driest (also the only year with a drought that ramped up to "severe" in September), the shortest, the latest SOS and EOS and had the highest thermal sum, also unlike our findings that growth increases with warmer seasons (Figure S1 and Figure S2).

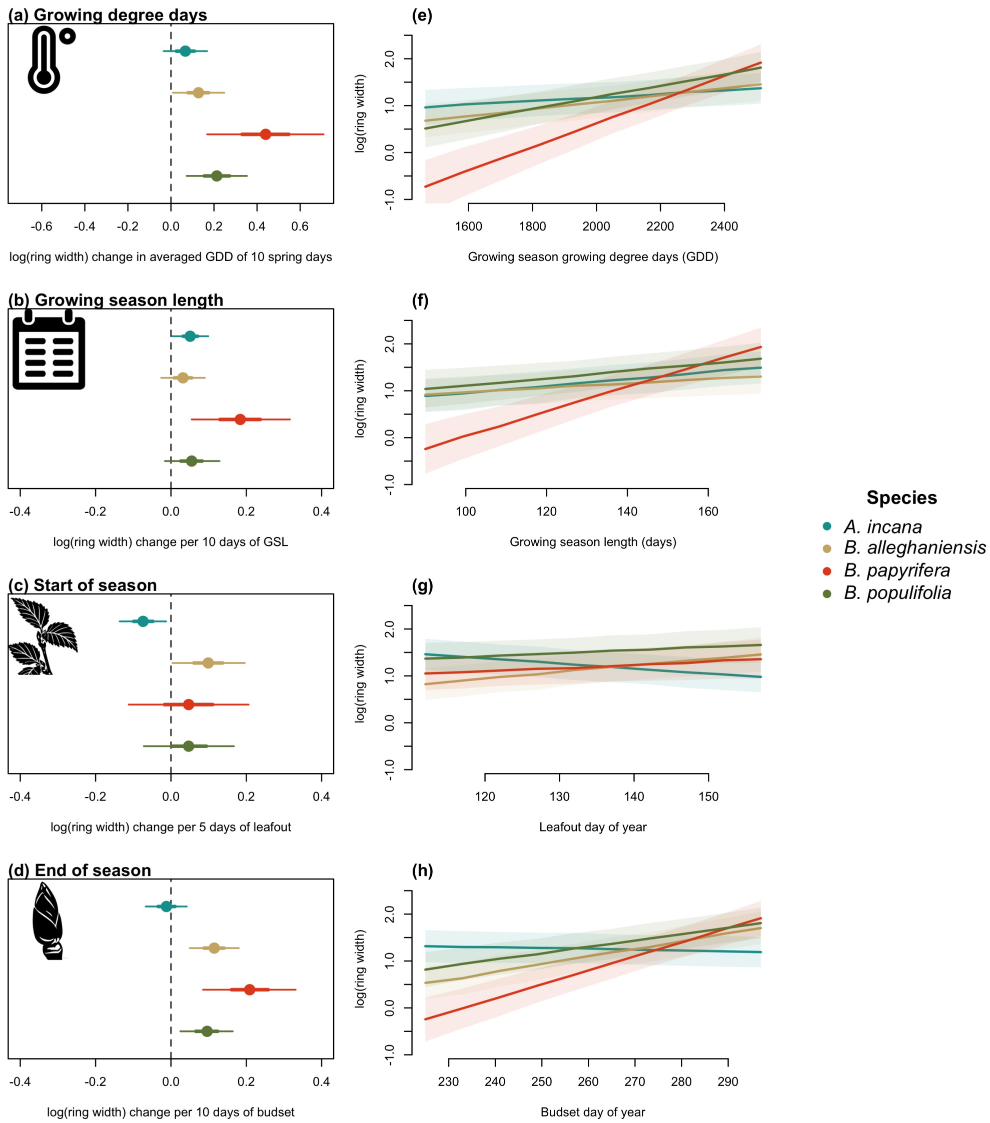

Figure 1: Model slope parameters across the different model predictors for Wildchrokie species. GDD represent the accumulated growing degree days between leafout and budset (a) the models estimates shows log ring width change per 7 average spring days GDD (174.05). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). All predictors were fit using the same model and number of observations, with only the independent variable changing. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI. *A. incana* and *B. papyrifera* grew more with longer calendar seasons, but only *A. incana* grew less with later leafout, but had no effect in response to budset. Interestingly, *B. papyrifera* did not grow more with earlier leafout, but did with a later budset and longer calendar season. *B. alleghaniensis* and *B. populifolia* did not grow more with longer calendar seasons, but *B. alleghaniensis* grew more with a later leafout and budset whereas *B. populifolia* only grew more with a later budset.

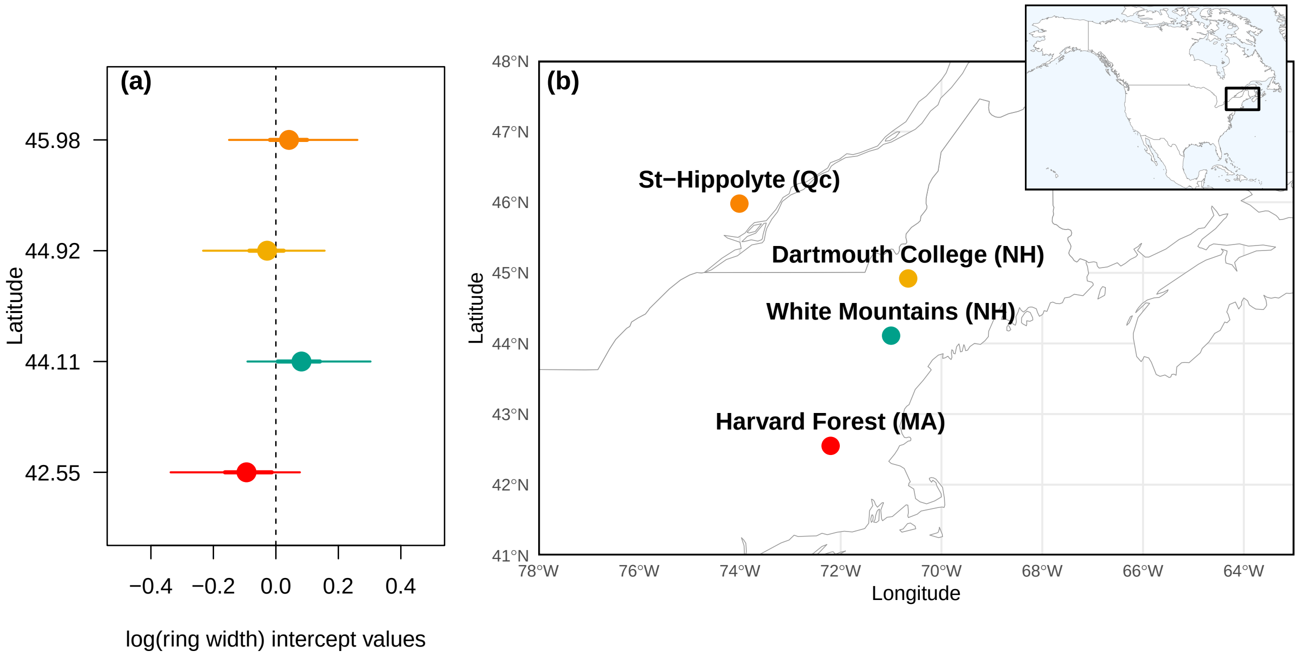

Figure 2: The effect of provenances on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI. The sites are color-coded from (b), map showing their locations.

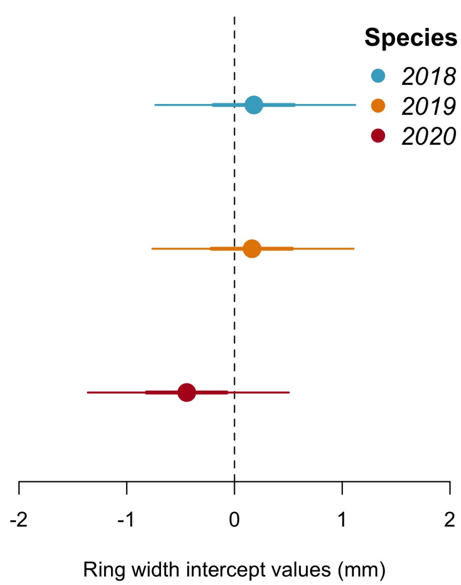

Figure 3: Model estimates for the intercepts on year. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Supplemental results

Table S1: Species level slope parameters (β and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	β =0.07, [-0.03, 0.17]	β =0.05, [0.00, 0.10]	β =-0.07, [-0.14, -0.01]	β =-0.01, [-0.07, 0.04]
B. alleghaniensis	β =0.13, [0.01, 0.25]	β =0.03, [-0.03, 0.09]	β =0.10, [0.00, 0.20]	β =0.12, [0.05, 0.18]
B. papyrifera	β =0.44, [0.17, 0.71]	β =0.18, [0.06, 0.32]	β =0.05, [-0.11, 0.21]	β =0.21, [0.09, 0.33]
B. populifolia	β =0.21, [0.07, 0.35]	β =0.06, [-0.02, 0.13]	β =0.05, [-0.07, 0.17]	β =0.10, [0.03, 0.16]

Table S2: Site level intercept parameters (α and 90% uncertainty intervals, in square brackets) for the GDD model.

Site	Intercept.estimate	Latitude	Longitude
	α =-0.03, [-0.23, 0.16]	44.79	-71.15
	α =-0.09, [-0.34, 0.08]	42.53	-72.19
	α =0.04, [-0.15, 0.26]	45.93	-74.03
	α =0.08, [-0.09, 0.3]	44.11	-71.23

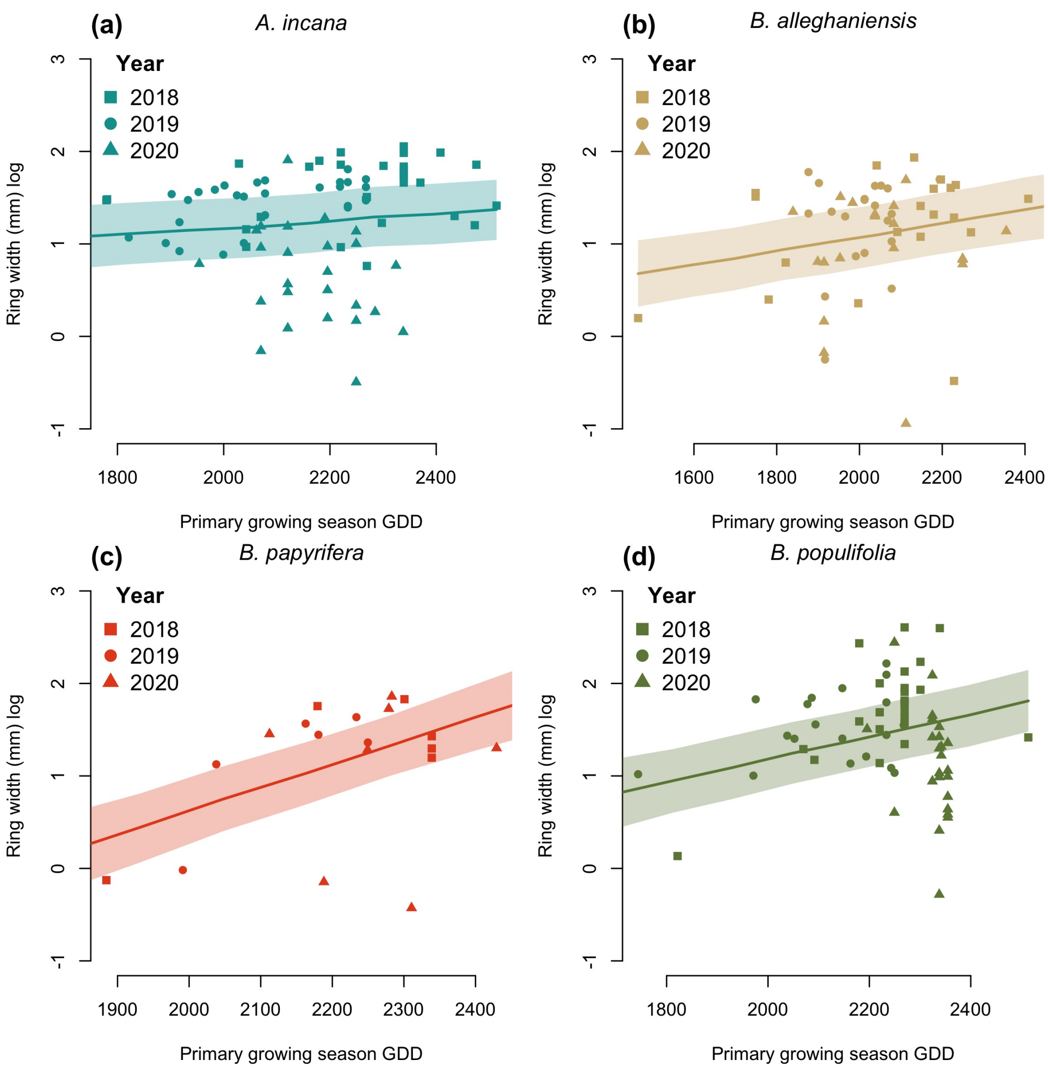

Figure S1: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

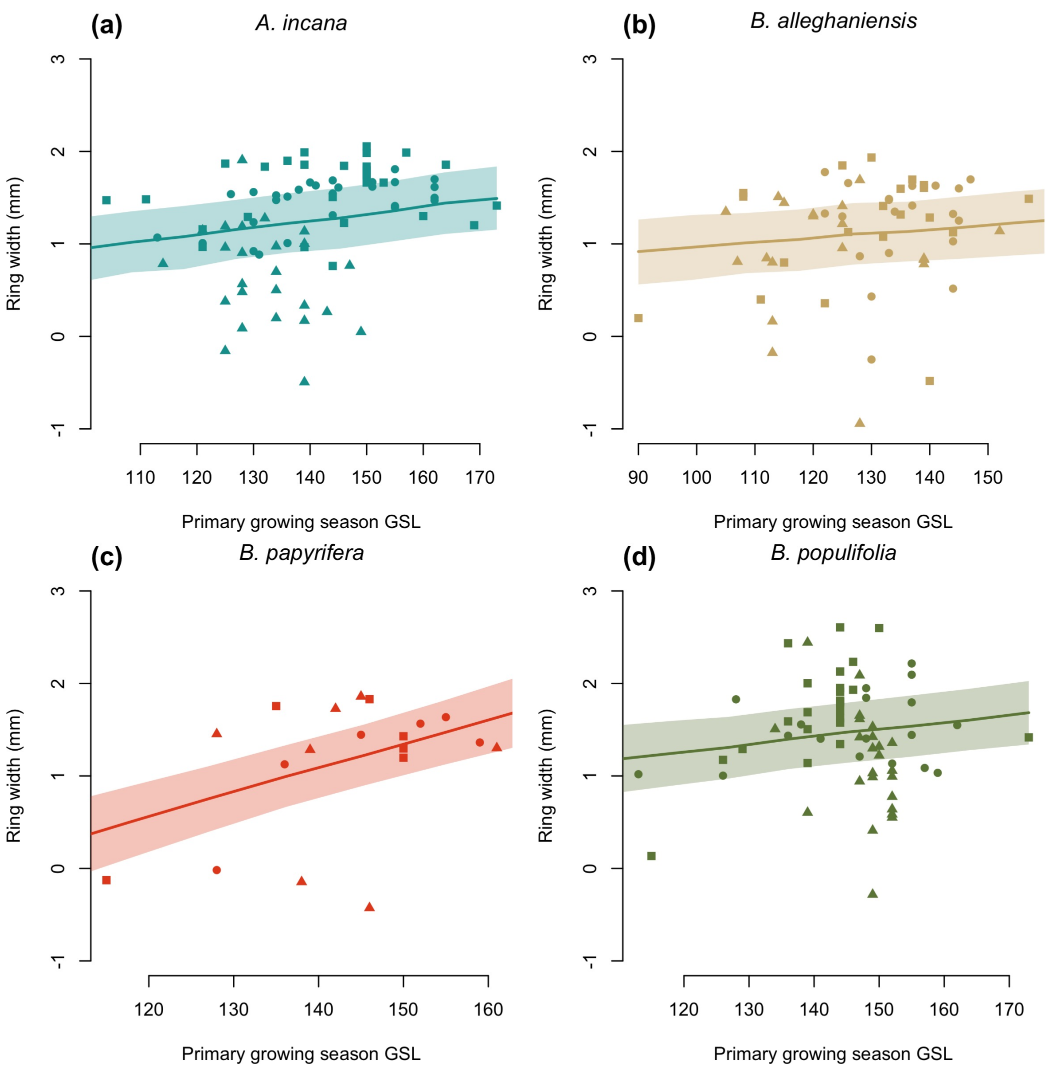

Figure S2: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

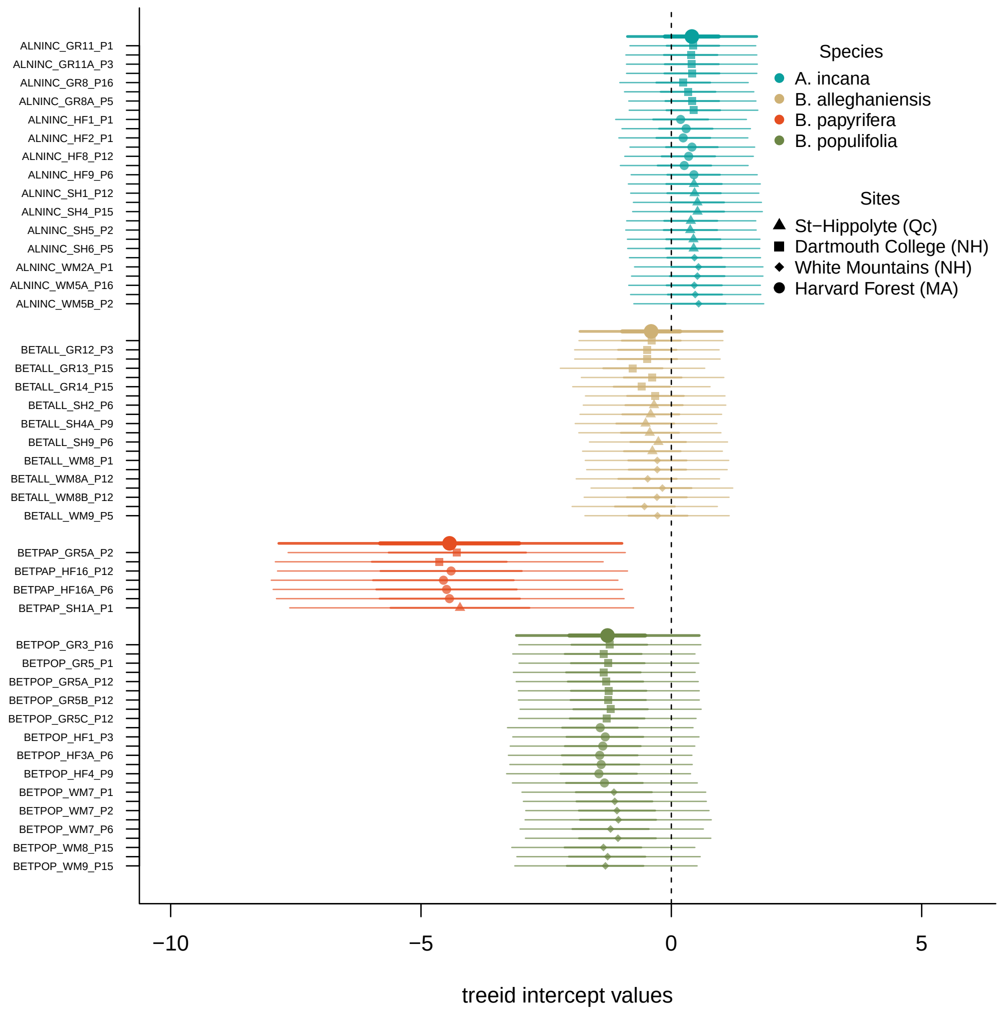

Figure S3: Intercept estimates for each group of the GDD model. The different treeid values represent the sum of the treeid, provenance, species and averaged year intercepts. The bigger dots and lines represent aggregated average across these values. the dots are the mean of the posterior distribution and the lines, the 50 and 90% UI.

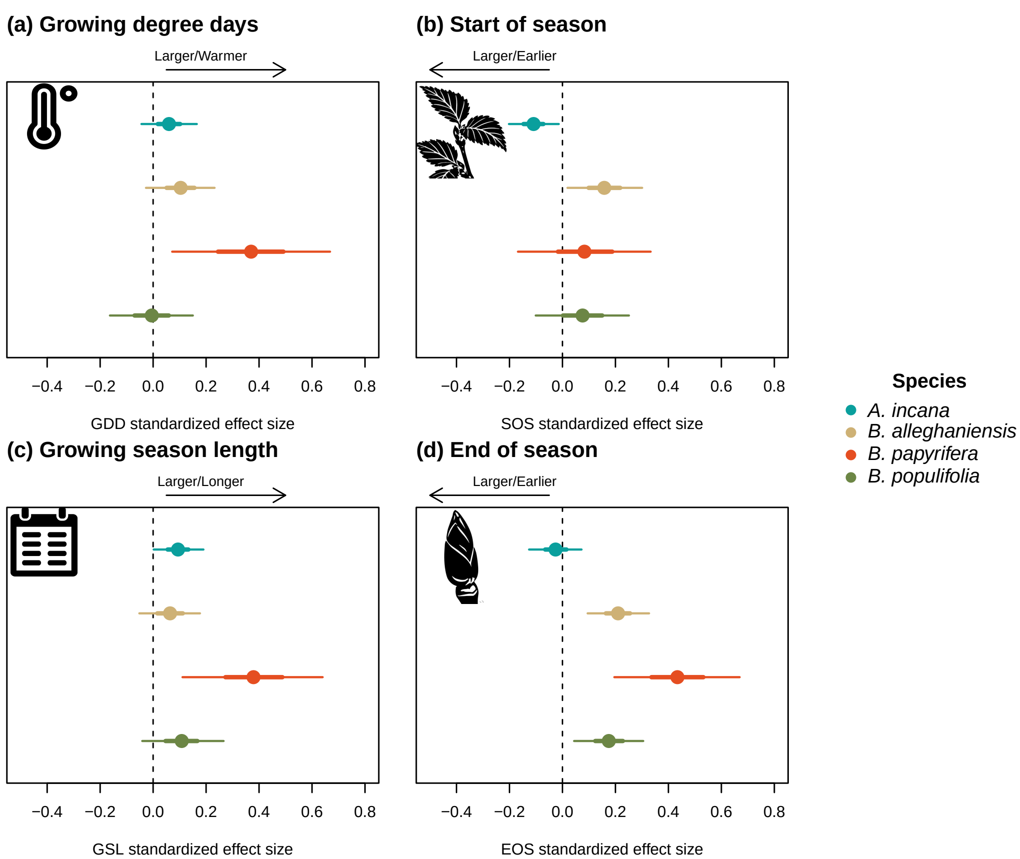

Figure S4: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

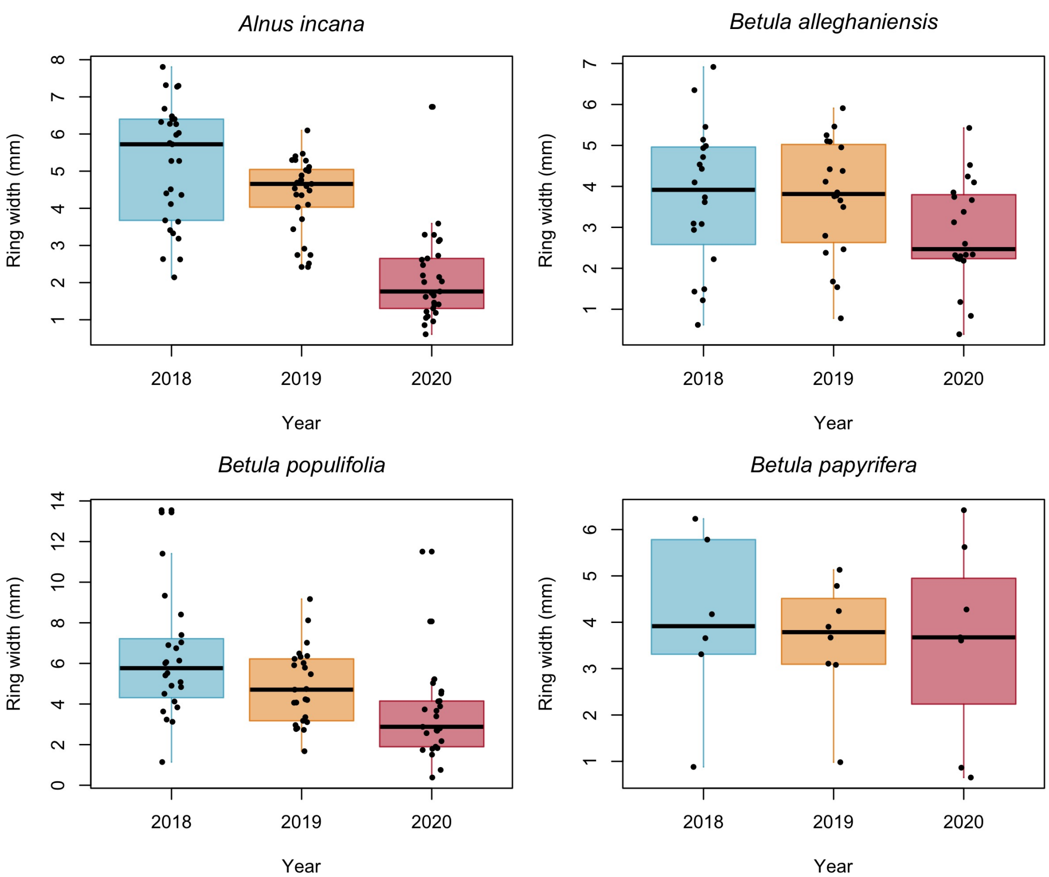

Figure S5: Growth data across years and species. The dots represent the observed ring widths. The boxes are color coded by years.